

Estimating Node Abnormalities From Imprecise Subgraph-Level Supervision

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Abstract—Although existing node anomaly detection techniques following supervised or unsupervised paradigms have achieved empirical success, the expensive annotation cost and the high possibility of spotting uninteresting outliers are still two major limitations. Interestingly, with limited time, workforce, and expertise, it seems easier for regulators to roughly screen out a suspicious region likely to contain anomalies rather than precisely locate the problematic individual. Thus, it is appealing to investigate how to learn node abnormalities from such coarse-grained imprecise subgraph-level supervision, related research however has received little scrutiny. In this paper, we generalize classic multiple instance learning to graph data and propose a novel architecture ASSESS which regards subgraphs as bags and nodes as bag instances. By pulling apart the score gap between anomalies and normal nodes through inter-bag loss and intra-bag loss, ASSESS tends to assign higher scores to abnormal nodes so that node anomalies can be detected. To further enhance the adaptability to low labeling rates, the self-training mechanism is introduced to ASSESS++, which automatically explores subgraphs that may cover abnormal nodes as pseudo-supervision by measuring the node score distribution in the subgraph. Experiments on real-world benchmark datasets corroborate the superiority of our proposed model *w.r.t.* AUC and AP.

Index Terms—Graph anomaly detection, weakly-supervised learning, multiple instance learning, graph neural networks.

I. INTRODUCTION

A VARIETY of complex interconnected systems in the real world, such as social networks, collaboration networks, and economic networks, can be abstracted as

graph-structured data¹ that describe the rich semantics and interactions of multiple entities through nodes and edges [2]. To extract valuable insights from such data represented in the form of graphs/networks, there has been a great deal of interest in many graph mining tasks, including node/graph classification [3], [4], [5], community detection [6], network partition [7], personalized recommendation [8], [9], [10], [11], and click-through rate (CTR) prediction [12], [13]. As one of the crucial tasks, node anomaly detection, which aims to uncover suspicious nodes that significantly deviate from the majority of instances in the graph, has attracted much attention recently [14], [15], [16] since it helps regulators identify the risks of detrimental events occurring in various realistic networks and take early action to control them [17]. For example, exposing potential fraudsters in financial networks is beneficial to maintaining the security and stability of economic systems [18]; controlling the spreaders of fake news in social networks can prevent them from creating panic and chaos in society [19]. No matter what kind of networks being detected, according to the adopted learning paradigm, existing techniques could be roughly divided into the following categories: *supervised/semi-supervised* [20] and *unsupervised* [21]. Although unsupervised approaches avoid laborious labeling costs, which are indispensable for supervised models that rely on large-scale annotations, they are prone to spot noisy or uninteresting outliers as anomalies without knowing what true anomalies look like [22]. The semi-supervised paradigm based on a small set of labeled samples seems to alleviate the above shortcomings. In practice, it may still suffer from the lack of fine-grained node anomaly labels due to insufficient domain knowledge and expert experience required for annotation.

Recall that in practical scenarios, regulators are expected to analyze data as quickly as possible and provide early warning of risks. Due to the diversity and complexity of anomalies, precise node-level labeling requires sufficient prior knowledge and detailed one-by-one analysis, which is inevitably time-consuming and labor-intensive. In contrast, it seems more convenient for regulators to spend much less effort screening out regions likely to cover anomalies [23]. For example, as shown in Fig. 1, during the investigation of financial fraud cases, law enforcement can easily delineate the scope of suspects through preliminary analysis, and fraudsters engaged in money laundering are highly likely to exist among them. However, it is significantly more difficult to pinpoint the real fraudsters in a short period of time due to intricate information or evidence [24]. Note that the above

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¹ It can also be referred to as network data [1].

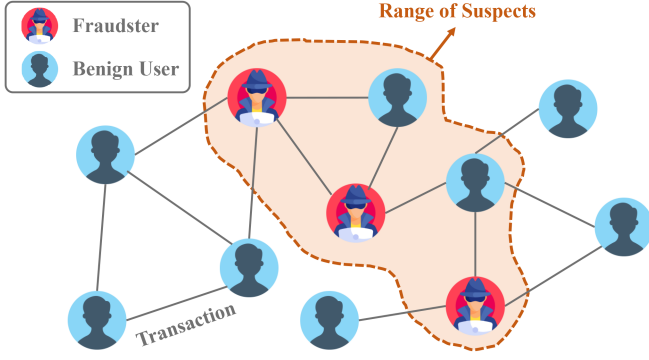


Fig. 1. A toy example of the range of potential suspects in a financial transaction network, which most likely contains anomalies (fraudsters).

coarse-grained annotations are typically presented in the form of subgraphs, and such subgraph-level labels contain minimal supervision provided to the model. So, here arises a question: *can we accurately identify node-level anomalies with only imprecise subgraph-level supervision?*

Academically, the above task belongs to the category of *weakly supervised learning* [25], which is extremely compelling for practical applications since coarse-grained subgraph-level supervisory signals are more easily accessible than precise node-level labels. The weakly labeled subgraph only reflects the presence of abnormal nodes somewhere in it, but the specific anomalies are still unknown, hence, it is called imprecise/inexact supervision. However, taking advantage of such coarse-grained subgraph-level weak supervision for node-level anomaly detection is not a trivial task. The following challenges exist here. First, previous works [26], [27], [28] generally formulate similar weakly supervised problems as multiple instance learning (MIL) [29]. Despite the empirical success, these methods are almost all oriented to grid-like image data in the computer vision domain and fail to capture the topological structures between instances. Moreover, they focus on group/bag-level prediction, such as image/object classification, rather than instance prediction within the bag. Thus, existing MIL methods cannot be directly used on graphs to spot fine-grained node anomalies. Second, imprecise abnormal subgraph labels are noise-prone as they include a mixture of benign and anomalous nodes. Thus, they cannot accurately provide the model with node-level supervision that distinguishes normal from abnormal. How to mitigate the negative impact of imprecise labeling and convert such subgraph labels into node-level constraints to capture the underlying anomaly patterns through modeling inter-subgraph differences poses another challenge. To the best of our knowledge, existing studies have put very little effort into weakly supervised node anomaly detection [23], which also motivates our further research.

To overcome the above challenges and promote the development of the related community, in this paper, we present a novel MIL architecture adapted to graph data, named ASSESS, for spotting node-level anomalies under imprecise subgraph-level supervision. Analogous to the classic MIL concept, we treat subgraphs as bags in which different nodes constitute instances

in a bag and have inherent topological relations rather than independence. Then, we rank nodes with the maximum scores in normal and abnormal subgraphs via the inter-bag loss, and simultaneously pull apart the difference between potential anomalies and normal nodes in the abnormal subgraph via the intra-bag loss. Together with the constraints specifically designed for graph-structured data and anomaly detection, ASSESS is more inclined to assign high scores to anomalies such that abnormal nodes can be detected. To further enhance the adaptability of the model to low labeling rates, the self-training [30] mechanism is introduced to form a variant ASSESS++. By automatically mining potential candidate subgraphs from the unlabeled set that are similar to the score distribution of existing labeled abnormal subgraphs to generate pseudo labels, ASSESS++ expands the available label set iteratively, which significantly alleviates the high cost required to obtain fine-grained annotations. Our main contributions are summarized as follows:

- We highlight the practical significance of studying weakly supervised node anomaly detection, which currently receives less attention and scrutiny, and generalize the classic MIL to the graph domain, providing a promising solution for this community.
- We present a novel MIL framework ASSESS to learn node abnormalities from imprecise subgraph-level supervision via inter- and intra-bag anomaly ranking while introducing ASSESS++ to automatically generate pseudo labels to adapt to low labeling rate cases.
- We empirically demonstrate the superiority of the proposed ASSESS and ASSESS++ in terms of AUC and AP on real-world benchmark datasets that involve multiple types of networks, and intuitively illustrate the detection results through visual analysis.

The rest of the paper is organized as follows. We start by providing a relatively comprehensive literature review on node-level anomaly detection and briefly introducing some related works on multiple instance learning in Section II. Then, in Section III, we give a formal definition of node anomaly detection under subgraph-level weak supervision and present our MIL based framework ASSESS and ASSESS++ in detail. Experimental evaluations on multiple benchmark datasets and discussions are placed in Section IV. Finally, Section V concludes the whole paper and visions the future work.

II. RELATED WORKS

In line with the focus of our work, we intend to provide a relatively comprehensive literature review on node-level anomaly detection in graphs and briefly review the previous works on multiple instance learning. Hopefully this will help general readers from different disciplines comprehend the background and motivation better.

A. Node-Level Anomaly Detection in Graphs

Background: In computer science, research on anomaly detection, *a.k.a.* outlier detection, can be traced back to the 1960s [31], which aims to detect rare, unusual, and suspicious ones that

TABLE I
SUMMARY OF RELATED WORKS ON NODE-LEVEL ANOMALY DETECTION IN GRAPHS

	Algorithm	Year	Form of Supervision	Learning Paradigm	Application Data
Non-Deep Learning	SCAN [39]	2007	None	Unsupervised	Political Book Graph, Customer Graph
	OddBall [40]	2010	None	Unsupervised	Citation Network, Email Network
	NrMF [41]	2011	None	Unsupervised	Author-Paper Graph, User-Movie Graph
	CODA [42]	2010	None	Unsupervised	DBLP Co-authorship Network
	ConSub [43]	2013	None	Unsupervised	Co-purchase Network, Email Network
	ConOut [44]	2014	None	Unsupervised	Co-purchase Network, DBLP Network
	ALAD [45]	2017	None	Unsupervised	Co-purchase Network, Citation Network
	Radar [46]	2017	None	Unsupervised	Co-purchase Network, Email Network
	ANOMALOUS [47]	2018	None	Unsupervised	Co-purchase Network, Email Network
	SemiGNN [48]	2019	Small Portion of Node Labels	Semi-Supervised	ALIPAY Multi-View Social Network
Deep Graph Learning	CARE-GNN [49]	2020	Small Portion of Node Labels	Semi-Supervised	Yelp and Amazon Review Network
	[50]	2021	Small Portion of Node Labels	Semi-Supervised	Co-purchase Network, Citation Network
	GDN [51]	2021	Few Node Labels	Few-Shot Learning	Review Network, Citation Network
	GCCAD [52]	2022	Sufficient Node Labels	Supervised	Review Network, Citation Network
	BWGNN [53]	2022	Sufficient Node Labels	Supervised	Social Network, Transaction Network
	AMNet [54]	2022	Small Portion of Node Labels	Semi-Supervised	Social Network, Transaction Network
	GHRN [55]	2023	Small Portion of Node Labels	Semi-Supervised	Social Network, Transaction Network
	CONSIGAD [56]	2024	Few Node Labels	Consistency Training	Social Network, Transaction Network
	GGAD [57]	2024	Small Portion of Labeled Normal Nodes	Semi-Supervised	Social Network, Transaction Network
	DOMINANT [58]	2019	None	Unsupervised	Co-purchase Network, Social Network
	AnomalyDAE [59]	2020	None	Unsupervised	Social Network, Citation Network
	GUIDE [60]	2021	None	Unsupervised	Citation Network
	CoLA [61]	2021	None	Unsupervised	Social Network, Citation Network
	ANEMONE [62]	2021	None	Unsupervised	Citation Network
	ComGA [63]	2022	None	Unsupervised	Co-purchase Network, Email Network
	CONAD [64]	2022	None	Unsupervised	Social Network, Email Network
	GRADATE [14]	2023	None	Unsupervised	Citation Network, Air-Traffic Network
	RAND [65]	2023	None	Unsupervised	Social Network, Citation Network
	TAM [66]	2023	None	Unsupervised	Social Network, Citation Network
	MDA-GAD [15]	2024	None	Unsupervised	Social Network, Citation Network
	ADA-GAD [67]	2024	None	Unsupervised	Social Network, Email Network
	GAD-NR [68]	2024	None	Unsupervised	Co-purchase Network, Social Network
	SmoothGNN [69]	2025	None	Unsupervised	Social Network, Transaction Network
	EAGLE [70]	2025	None	Unsupervised	Citation Network, Transaction Network
	WEDGE [23]	2023	Small Portion of Subgraph Labels	Weakly Supervised	Review Network, Citation Network

deviate significantly from the majorities in a dataset [17], [18]. Many useful techniques have been developed over the past decades, especially for identifying anomalies in a set of multidimensional data points [32], [33] such as well-known LOF [34] and Isolation Forest [35] since conventional methods typically represent real-world entities as multidimensional feature vectors (e.g., texts posted by users in social media is represented as bag-of-words) [17]. Despite their empirical success, they all inadvertently ignore potential associations between the data. In fact, many entities in reality are inherently interconnected and exist in the form of networks. For example, the social relationships between people constitute a social network; the physical associations between proteins in a cell or organism form a PPI biological network. Fortunately, graphs, in which nodes denote entities and edges characterize their relationships, have proven to be a powerful tool to effectively model and analyze various networks. As a result, graph anomaly detection, which aims to find suspicious graph objects (including nodes, edges, subgraphs) that differ significantly from the majority of the reference objects in the graph [18], has begun to attract considerable and increasing attention from researchers. Compared to other graph objects, current research on graph anomaly detection mostly focuses on node-level anomaly detection [17], which has broad applications in many high-impact domains, such as social media [36], finance [37], and healthcare [38]. Next, we will concentrate on reviewing previous works on node-level anomaly detection to provide a research context that

is strongly relevant to this paper. A brief summary is shown in Table I.

Methodology: In the early stage, plain graphs containing only topological information were often used to describe various complex networks. Many detection methods exploit structural statistical properties such as node degree or subgraph centrality to find patterns and spot anomalies. For instance, SCAN [39] clusters nodes based on a structural similarity measure to expose hubs and outliers in the graph. OddBall [40] leverages statistical features extracted from each node and its one-hop neighbors to detect specific structural anomalies. Additionally, NrMF [41] attempts to employ matrix factorization to mine the community structure in the adjacency matrix to identify anomalies. As attributed graphs (or simply called graphs) carrying node feature information are now widely used to model realistic interconnected systems, most existing methods are absorbed in processing and analyzing such graph data. Among them, many early methods spot anomalies by analyzing the congruence between attributes and topology [42], [43], [44]. For instance, CODA [42] employs a probabilistic model to characterizes both attributes and edge links simultaneously by defining their joint distribution. Driven by the good ability of matrix factorization in jointly modeling attributes and topology, various matrix factorization based methods have been put forward [45], [46], [47]. For instance, Radar [46] spots anomalies by modeling the residuals of attribute information and its coherence with network information. ANOMALOUS [47] jointly conducts anomaly

detection and attribute selection based on CUR decomposition and residual analysis. However, these traditional machine learning techniques still have limitations in seamlessly integrating graph topology and node attributes, and are difficult to adapt to large-scale graphs.

Benefiting from the outstanding performance of deep learning, methods based on various graph neural networks (GNN), such as graph convolutional networks (GCN) and graph attention networks (GAT) [2], have become the current mainstream. Depending on the learning paradigm adopted, existing deep models can be broadly classified into supervised/semi-supervised, unsupervised, and weakly supervised. Supervised/semi-supervised methods [48], [49], [50], [52], [54] focus on leveraging labeled normal and anomalous samples to learn pattern boundaries of both categories to detect anomalies. For example, BWGNN [53] utilizes a supervised learning approach where Beta graph wavelets are used to generate band-pass filters. GGAD [57] spots anomalies in a semi-supervised manner by generating pseudo abnormal nodes to provide effective negative node samples for training a discriminative one-class classifier under the condition that only a small portion of labeled normal nodes are accessible. These methods, however, heavily depend on the availability of labeled data, which limits their scalability and applicability in real-world scenarios where labeled data is scarce. In contrast, due to the advantage of not relying on expensive manual labeling, the vast majority of existing works [15], [61], [63], [64], [65], [68], [70] fall under the unsupervised category. The traditional detection methods mentioned above also belong to this category. To effectively capture the difference between abnormal and benign without labels, existing deep models mainly adopt two strategies: reconstruction and contrastive learning. The first one calculates the residuals between the input data and its reconstructed version estimated by the model to detect anomalies as content patterns of anomalies deviate significantly from the majority and are difficult to reconstruct. An early attempt at this deep method is DOMINANT [58]. It introduces a GCN based deep autoencoder to reconstruct the input graph and flags nodes with high residual errors as anomalies. As an improved version of DOMINANT, AnomalyDAE [59] employs a dual autoencoder to encode the graph from structural and attribute perspectives respectively for better reconstruction quality. ComGA [63] further considers the community structure during anomaly detection by designing a tailored GCN to encode the modularity matrix. Contrastive learning based methods capture the pattern differences between normal and abnormal by learning to distinguish between positive and negative sample pairs. The first attempt was CoLA [61], which makes the target node and its local subgraph as a contrastive pair (*i.e.*, node-subgraph contrast), and spots anomalies by learning and evaluating the consistency between the node and its local neighboring substructure. On this basis, ANEMONE [62] adds node-node contrast through a multi-scale contrastive learning objective, and GRADATE [14] further considers subgraph-subgraph contrast between the original input graph and the augmented graph modified by edge perturbation. Some methods combine reconstruction and contrastive learning to finely expose the discrepancy between abnormal and benign ones. For example, CONAD [64] models prior human knowledge by data augmentation and integrates it into the encoding

process through a well-designed contrastive loss, then discerns anomalies based on reconstruction errors. MDA-GAD [15] employs a hybrid learning scheme including graph reconstruction and multi-view discrimination to identify anomalies. While the unsupervised methods are more scalable and practical, they are prone to spot noisy or uninteresting anomalies and suffer from undesirable detection accuracy since they do not know what anomalies the user really cares about.

Compared to the impracticality of obtaining fully labeled datasets, coarse-grained annotations that typically require marginal time and labor costs are relatively accessible in practical scenarios. Studying weakly supervised node-level anomaly detection with coarse-grained supervision help alleviate the drawbacks of the above supervised and unsupervised methods. However, related research is still in its infancy. One notable work is WEDGE [23], which spots critical node-level anomalies with an attention mechanism and contrastive learning under a small portion of coarse-grained subgraph labels. To promote related research, we propose a novel weakly supervised model ASSESS, which detects node anomalies under the subgraph-level supervision by pulling apart the score gap between anomalies and normal nodes through inter-bag loss and intra-bag loss. ASSESS++ further enhances the adaptability to low labeling rates. The differences between WEDGE and ours are discussed in detail in Section III-C.

Application: As shown in Table I, most existing models are not developed for a specific network but focus on designing general learning algorithms to be applied to anomaly detection scenarios in various networks such as Email networks, social networks, and transaction networks. For instance, in the Email network dataset Enron [43], where nodes represent Email addresses and the email transmission between them form edge links. Each node is described by 20 attributes such as average content length, different encodings count, and the average number of recipients. The email addresses having sent spam messages are regarded as anomalies and should be identified by the model. Take the social network dataset Weibo [21] as another example, where nodes indicate users registered on Tencent-Weibo and users who use the same hashtag are connected by edges. The attributes affiliated to each user consist of a one-hot vector describing the location of the post and the bag-of-words vector of the post. Users who engaged in at least five suspicious posting activities are considered anomalies that will be detected by the model. Generally speaking, in engineering practice, we can construct network data describing various complex interconnected systems based on detection requirements, and apply these detection models to discover anomalies for network risk warning and management.

B. Multiple Instance Learning

Multiple Instance Learning (MIL) is a technique that overcomes the limitations of traditional supervised learning when individual instance labels are unavailable [95]. It is a form of weakly-supervised learning under inexact supervision [25], where instances are organized in bags with coarse-grained labels. Formally, in MIL, a bag $B = \{x_1, x_2, \dots, x_n\}$ is labeled positive if at least one instance in the bag is positive (Y), otherwise, it is marked as negative (N). The objective in classic

TABLE II
SUMMARY OF RELATED WORKS ON MULTIPLE INSTANCE LEARNING

Algorithm	Year	Application Task
DMIL [71]	2015	Image Classification and Annotation
WSSM [72]	2016	Video Classification
[73]	2018	Video Anomaly Detection
C-MIL [74]	2019	Object Detection
[75]	2019	Cancer Image Classification
TCN-IBL [76]	2019	Video Anomaly Detection
MIL-GNN [77]	2019	Image and Text Classification
D-MILN [78]	2020	Document Sentiment Classification
FS-GCN-MIL [79]	2020	Histopathological Image Classification
DSMIL [80]	2021	WSI Classification
MI-AOD [81]	2021	Object Detection
D-MIL [82]	2022	Object Detection
Mutual-Att [83]	2022	Offensive Language Detection
NAGCN [84]	2022	WSI Classification
MHIM-MIL [85]	2023	WSI Classification
NG-MIL [86]	2023	Video Anomaly Detection
WEDGE [23]	2023	Node-Level Anomaly Detection
CaMIL [87]	2024	WSI Classification
INS [88]	2024	WSI Classification
LaFiCMIL [89]	2024	Large File Classification
IMPOMIL [90]	2024	Medical Image Classification
PTG [91]	2024	3D MRI Image Classification
[92]	2024	Drug Activity Prediction
[93]	2024	Video Anomaly Detection
IFS-VAD [94]	2025	Video Anomaly Detection

MIL is to learn a mapping function $f : \mathcal{B} \rightarrow \mathcal{Y}$ from a training set $T = \{(B_1, y_1), \dots, (B_k, y_k)\}$, where $B_i \in \mathcal{B}$, and $y_i \in \mathcal{Y} = \{Y, N\}$ is the bag-level label. The goal is to predict labels for unseen bags according to the learned f . Typically, depending on how the information in multi-instance data is exploited, MIL can be divided into two paradigms [96]. In the Bag-Space (BS) paradigm, the discriminative learning process occurs at the bag-level. The classifier makes use of the information from the whole bag B (e.g., bag embeddings and scores) to establish discrimination between bags. In contrast, the Instance-Space (IS) paradigm views discriminative information at the instance-level. The classifier separates the instances in positive bags from those in negative bags by considering the characteristics of individual instances.

MIL has been successfully applied to multifarious real-world scenarios [97]. Table II presents a brief summary of related works. In the computer vision domain, DSMIL [80] incorporates self-supervised contrastive learning and multiscale features into the MIL model for whole slide image (WSI) classification. D-MIL [82] employs multiple MIL learners to explore discrepant but complementary solutions indicating object parts, which are then fused with a collaborative module to achieve precise object localization. WSSM [72] combines MIL and conditional random field techniques seamlessly to model sequence data for video classification. As for text processing and analysis, D-MILN [78] achieves document-level sentiment classification by learning aspect-level classifiers from the back propagation of document-level supervision. Mutual-Att [83] presents a hybrid fusion MIL model with mutual-attention mechanism to detect offensive language with document-level supervision. In the medical field, [75] considers each patient as a “bag” labeled with their diagnosis for histopathological image analysis. PTG [91] is a patch-target guided dual-branch MIL framework that explores

local features and significant regions simultaneously for 3D MRI data analysis. Note that there have been several studies on MIL-based methods in the field of video anomaly detection [73], [76], [86], [93]. For instance, [73] introduces a MIL ranking loss with sparsity and smoothness constraints to spot anomalous video segments with video-level weak supervision. [93] further proposes a prompt-enhanced MIL framework to detect various abnormal events in videos while ensuring clear event boundaries. However, these methods are inapplicable to graph data due to the unique topological relationships between nodes. Although some efforts, such as MIL-GNN [77], FS-GCN-MIL [79], and NAGCN [84], combine graph neural networks with MIL, they focus on capturing potential relationships between instances with GNN to solve tasks like weakly supervised image classification or text categorization. Research that directly targets mining and analysis tasks on natural graph-structured data, such as node-level anomaly detection, is very scarce. A notable work is the previously mentioned WEDGE, which follows the Bag-Space paradigm to uncover node anomalies by enforcing distinctive representations between normal and abnormal subgraphs. To expand the application of MIL in graph data mining, we propose a novel MIL framework tailored for graph data to identify node-level anomalies under imprecise subgraph-level supervision, which follows Instance-Space paradigm to establish discriminative information at the node instance level by inter-bag and intra-bag ranking objectives.

C. Summary

As can be seen from Section II-A and Table I, the research on how to exploit imprecise subgraph-level supervision to identify anomalies in weakly supervised node anomaly detection is still in its initial stage, although there are many application scenarios where coarse-grained labeling is available in reality. However, as a promising solution to imprecise supervision, Section II-B and Table II illustrate that existing MIL attempts for analysis tasks on graph data are very limited. In this sense, our studied MIL framework tailored for weakly supervised node anomaly detection in graphs is of great significance. Moreover, compared to WEDGE that follows the Bag-Space paradigm, our modeling strategy is more fine-grained since ASSESS establishes discriminative information at the instance-level. ASSESS++ further enhances the adaptability to low labeling rates, exhibiting good engineering application value.

III. METHODOLOGY

This section begins with a formal definition of our studied problem. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ denote a graph with $|\mathcal{V}| = n$ nodes and $|\mathcal{E}| = m$ edges, where each node $v_i \in \mathcal{V}$ is affiliated with $|\mathcal{F}| = d$ attributes. A subgraph in $\mathcal{G}$ is defined as $\mathcal{S} = (\mathcal{V}_S, \mathcal{E}_S, \mathcal{F})$, where $\mathcal{V}_S \subseteq \mathcal{V}$ and $\mathcal{E}_S \subseteq \mathcal{E} \cap \mathbb{P}(\mathcal{V}_S)$. $\mathbb{P}(\mathcal{V}_S)$ is the powerset of $\mathcal{V}_S$. Analogous to the definition of MIL, in our setting, a subgraph $\mathcal{S}_i = \{v_1^{(i)}, v_2^{(i)}, \dots, v_{b_i}^{(i)}\}$ is considered as a bag of size b_i , where different nodes make individual instances in the bag. If a subgraph contains at least one anomalous node, we call it an abnormal subgraph (positive bag) and mark it as $\mathcal{S}^a$. Otherwise, it is called a normal subgraph (negative bag) and

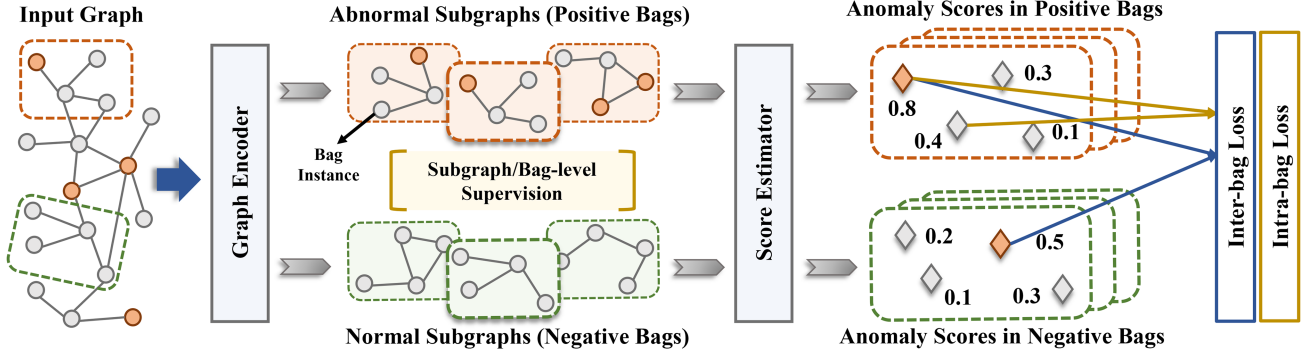


Fig. 2. Illustration of the proposed MIL framework ASSESS, which regards subgraphs as bags and nodes as bag instances. By modeling inter-bag and intra-bag anomaly ranking, ASSESS uncovers node anomalies under subgraph-level weak supervision.

denoted as $\mathcal{S}^n$. Assume that we are given a set of k labeled abnormal subgraphs $\mathcal{C}^a = \{\mathcal{S}_1^a, \mathcal{S}_2^a, \dots, \mathcal{S}_k^a\}$, then the set of nodes included in $\mathcal{C}^a$ is denoted as $\mathcal{V}^l$, and the remaining unlabeled node set is $\mathcal{V}^u$. Note that $\mathcal{V} = \{\mathcal{V}^l, \mathcal{V}^u\}$ and $|\mathcal{V}^l| \ll |\mathcal{V}^u|$ as only a few labeled abnormal subgraphs are available in advance. Following the convention of graph anomaly detection [18], our studied problem is formulated as a ranking task, which is formally defined as follows:

Problem 1. Node Anomaly Detection under Subgraph-Level Weak Supervision: Given a graph $\mathcal{G}$ represented by the adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and attribute matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, as well as a set of k labeled abnormal subgraphs $\mathcal{C}^a$. The task aims to learn a node-level anomaly detection model that is able to leverage knowledge from imprecise subgraph-level supervision $\mathcal{C}^a$ to learn the mapping from nodes to anomaly scores. Typically, abnormal nodes should be assigned larger scores and ranked higher than normal ones.

As can be seen, different from the above-mentioned classic MIL task which aims to predict labels for unseen bags, our studied problem is more challenging due to the following two aspects: (1) unlike the *i.i.d.* assumption on which most existing MIL works in the computer vision field is based, instances as nodes in bags in the form of subgraphs have inherent topological relations that must be explicitly considered; (2) our goal is to exactly uncover fine-grained anomalous nodes (instance-level prediction) under the guidance of subgraph-level weak supervision, rather than coarse-grained bag-level prediction.

A. Learning From Subgraph-Level Labels: ASSESS

Here we elaborate on the proposed framework ASSESS, which learns to identify node anomalies under the imprecise subgraph-level supervision. The overview of its architecture is illustrated in Fig. 2.

1) **Node Representation Modeling:** As for the input graph $\mathcal{G}$, we first need to take a general graph neural network (GNN) as a graph encoder to convert nodes into informative embeddings $\mathbf{H} = \text{GNN}(\mathbf{A}, \mathbf{X}) = \{\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n\} \in \mathbb{R}^{n \times h_1}$ for quantitative anomaly assessment. The graph encoder here is compatible with arbitrary GNN architectures [98], [99], [100]. Hidden vectors $\mathbf{H}$ are then projected into another latent space to obtain the

representation $\mathbf{Z} \in \mathbb{R}^{n \times h_2}$ through a nonlinear transformation:

$$\mathbf{Z} = \sigma(\mathbf{H}\mathbf{W}_t), \quad (1)$$

where $\mathbf{W}_t \in \mathbb{R}^{h_1 \times h_2}$ is a learnable parameter matrix, and σ denotes the parametric ReLU (PReLU) function [101]. Obviously, fully capturing the underlying semantic information from the input graph to derive high-quality node representations benefits the accurate estimation of abnormalities. Here, without external precise supervision, we draw on the idea of maximizing graphical mutual information in [102] to guide representation learning:

$$I(\mathbf{z}_i; \mathcal{G}_i) = I(\mathbf{z}_i; \mathbf{x}_i) + \sum_{j \in \mathcal{N}(i)} \frac{I(\mathbf{z}_i; \mathbf{x}_j)}{|\mathcal{N}(i)|} + I(\mathbf{w}_{ij}; \mathbf{A}_{ij}), \quad (2)$$

where $\mathcal{G}_i$ is the computation graph of node v_i that specifies the neighborhood around a particular node to perform a localized convolution. $\mathcal{N}(i)$ includes the neighbors of v_i , and $\mathbf{w}_{ij} = \sigma(\mathbf{z}_i^\top \mathbf{z}_j)$ with the sigmoid function σ . $I(\cdot)$ denotes the mutual information operator. Let the empirical probability distribution of node attributes $\mathbf{x}_i$ be $p(\mathbf{x}_i)$, the probability distribution of node representations $\mathbf{z}_i$ be $p(\mathbf{z}_i)$, and the joint distribution be $p(\mathbf{z}_i, \mathbf{x}_i)$. According to the information theory, the mutual information between $\mathbf{z}_i$ and $\mathbf{x}_i$ is defined as

$$I(\mathbf{z}_i; \mathbf{x}_i) = \int_{\mathcal{X}} \int_{\mathcal{Z}} p(\mathbf{z}_i, \mathbf{x}_i) \log \frac{p(\mathbf{z}_i, \mathbf{x}_i)}{p(\mathbf{z}_i)p(\mathbf{x}_i)} d\mathbf{z}_i d\mathbf{x}_i. \quad (3)$$

By minimizing $\ell_{rep} = -\sum_{v_i \in \mathcal{V}} I(\mathbf{z}_i; \mathcal{G}_i)$ over all node vectors, the learned representations $\mathbf{Z}$ finely preserve the respective attributes and topological semantics of nodes, laying the foundation for accurate anomaly estimation.

2) **Anomaly Ranking Modeling:** Given the final representations $\mathbf{Z}$, we define a score estimator to calculate real-valued anomaly scores for each node:

$$\mathbf{c} = \sigma(\mathbf{Z}\mathbf{w}_s), \quad (4)$$

where $\mathbf{w}_s \in \mathbb{R}^{h_2 \times 1}$ is a trainable scoring vector and the sigmoid activation σ we employed here aims to derive a reasonable anomaly score within the range of (0,1).

Considering that the node-level anomaly labels are unknown during training, we cannot learn the model simply by encouraging anomalies to score higher than normal nodes. Inspired by the maximum margin idea of classic MIL [73], [103], [104], we extend the notion of a margin from individual patterns to sets of patterns and design the following ranking loss with respect to the maximum scored node (instance) in the subgraph (bag):

$$\ell(\mathcal{S}_i^a, \mathcal{S}_i^n) = \max \left(0, 1 - \max_{v_p^{(i)} \in \mathcal{S}_i^a} c_p + \max_{v_j^{(i)} \in \mathcal{S}_i^n} c_j \right). \quad (5)$$

As can be seen, (5) enforces ranking only on two nodes with the highest scores in the abnormal subgraph $\mathcal{S}_i^a$ and the normal subgraph $\mathcal{S}_i^n$, respectively². Thus, it is referred to as inter-bag loss. Intuitively, node with the highest score in $\mathcal{S}_i^a$ is most likely to be the true anomaly, while node with the highest score in $\mathcal{S}_i^n$ is most likely a hard negative sample, which seems to be abnormal but is actually benign. We push the above two types of nodes far apart in terms of anomaly scores, which contributes greatly to achieving high anomaly detection accuracy. Recall that there is no pre-labeled normal subgraph $\mathcal{S}_i^n$ in our task setting. For efficiency, we randomly select k subgraphs with similar sizes to $\mathcal{S}_i^a$ from the unlabeled node set $\mathcal{V}^u$ as normal ones, leading to a set $\mathcal{C}^n$. Due to the rarity of anomalies, the sampled subgraphs have a high probability of being normal. Experiments later also validate the effectiveness of this strategy.

Note that $\mathcal{S}_i^a$ may not only have more than a single anomalous node but also contain normal nodes with a high probability, according to the definition of the annotated abnormal subgraph. Thus, we need to introduce additional ranking constraints to further optimize the solution space derived from (5). Specifically, we pick out a set of nodes with scores lower than the average anomaly score in $\mathcal{S}_i^a$, denoted as $\bar{\mathcal{S}}_i^a$. Nodes with a score above the average are more likely to be anomalies, while nodes in $\bar{\mathcal{S}}_i^a$ are more likely to be normal. We push them far away to enhance the detection model to distinguish normal nodes in the abnormal subgraph by the following intra-bag loss:

$$\ell(\mathcal{S}_i^a) = \max \left(0, 1 - \max_{v_p^{(i)} \in \mathcal{S}_i^a} c_p + \max_{v_q^{(i)} \in \bar{\mathcal{S}}_i^a} c_q \right). \quad (6)$$

The above-mentioned inter-bag loss and intra-bag loss are crucial for distinguishing meaningful anomalies from benign nodes under weak supervision. Specifically, in (5), through the max operation of term $\max_{v_p^{(i)} \in \mathcal{S}_i^a} c_p$, we define the margin of a positive bag (abnormal subgraph) with the margin of the “most positive” pattern (most likely anomaly). Besides, we find the “least negative” pattern (the hard negative pattern) in a normal subgraph through term $\max_{v_j^{(i)} \in \mathcal{S}_i^n} c_j$ and regard it as the margin of a negative bag. Once the margin between the positive and negative bags (*i.e.*, the margin between the anomaly and the indistinguishable normal) is pulled apart to form a distinguishing boundary through (5), the relative position of other sample

patterns in negative bags with respect to the boundary becomes irrelevant [103], and they can all be distinguished from abnormal. We also establish a boundary for the “most positive” pattern and the most likely negative pattern in a positive bag through (6) to further separate the normal mixed in weakly supervised subgraphs. You could refer to the paper [103] to better understand the maximum bag margin formulation of MIL.

3) *Sparsity and Homophily Constraints*: Two related factors should also be considered to capture anomalies effectively. For one thing, anomalies are usually rare in real-world scenarios, which is an important principle in anomaly detection tasks. In this sense, the anomaly scores of nodes in $\mathcal{S}_i^a$ should be sparse, *i.e.*, most tend to 0, since only a limited number of nodes may be abnormal. Formally, we minimize $\ell_{sp}(\mathcal{S}_i^a) = \sum_{v_p^{(i)} \in \mathcal{S}_i^a} c_p$ for each abnormal subgraph. For another, due to the topological nature of graphs, the desired scoring scheme should follow the Homophily assumption [105] and keep the scores of topologically connected nodes as close as possible. Thus, we enforce the Homophily constraint for each subgraph $\mathcal{S}_i$ (whether normal or abnormal) by minimizing the difference of anomaly scores for neighboring nodes, *i.e.*, $\ell_{hom}(\mathcal{S}_i) = \mathbf{c}_{\mathcal{S}_i}^T \mathbf{L}_{\mathcal{S}_i} \mathbf{c}_{\mathcal{S}_i}$, where $\mathbf{L}_{\mathcal{S}_i}$ denotes the Laplacian matrix of $\mathcal{S}_i$ and $\mathbf{c}_{\mathcal{S}_i}$ is the score vector of nodes in $\mathcal{S}_i$.

By integrating the above losses, we arrive at the complete objective function as follows:

$$\ell = \ell_{rep} + \ell(\mathcal{S}_i^a, \mathcal{S}_i^n) + \lambda_1 \ell(\mathcal{S}_i^a) + \lambda_2 \ell_{sp}(\mathcal{S}_i^a) + \lambda_3 \ell_{hom}(\mathcal{S}_i), \quad (7)$$

where multiple λ s are trade-off parameters that balance the importance of the above aspects. Since anomalies usually exhibit distinct patterns than normal nodes, the scores derived by $\mathbf{w}_s$ will be different. To discriminate abnormalities, $\ell(\mathcal{S}_i^a, \mathcal{S}_i^n)$ and $\ell(\mathcal{S}_i^a)$ constrain the score of anomalies to be greater than normal nodes. Furthermore, the sparsity and Homophily constraints make only a few nodes that may be anomalies have high scores (the situation where a normal node is assigned the largest score while an anomalous node is assigned a small score, which violates the sparsity constraint, will not occur). All the above ensures that the model is optimized in the right direction. The key steps of our proposed ASSESS are summarized in Algorithm 1.

B. Automatic Pseudo-Label Generation: ASSESS++

Although the weakly supervised setting weakens reliance on fine-grained node-level annotations, the limited time, prior knowledge, and labor costs may still prevent us from obtaining sufficient coarse-grained subgraph-level labels. Experiments on detection performance under various numbers of annotated abnormal subgraphs also reflect that too few labels will significantly restrict the model’s ability to uncover anomalies (see Fig. 8).

To improve the adaptability of ASSESS in practical cases with low labeling rates, we additionally incorporate a self-training module to create a variant version named ASSESS++. The core issue lies in how to select subgraphs that are more likely to contain anomalies from the huge unlabeled node space to generate pseudo labels. We analyze the scores assigned to nodes in normal

²This involves the conversion of the node index from a subgraph to the full graph. For brevity, the index of the score c follows the subscript of the node in the subgraph.

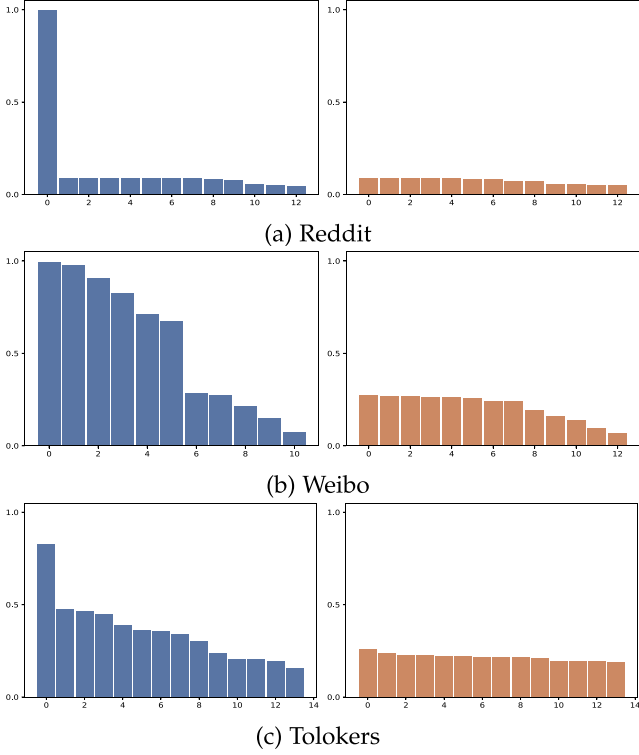


Fig. 3. The estimated anomaly scores for nodes in a randomly selected abnormal (left) and normal (right) subgraph.

Algorithm 1: The Learning Algorithm of ASSESS.

Input: Graph $\mathcal{G} = (\mathcal{A}, \mathcal{X})$, a set of k labeled abnormal subgraphs $\mathcal{C}^a$, parameters $\lambda_1, \lambda_2, \lambda_3$, training epochs E and batch size $2N$.

Output: Anomaly ranking of nodes in $\mathcal{G}$ (reverse order).

- 1: Construct $\mathcal{C}^n$ by randomly selecting k subgraphs from $\mathcal{V}^u$ as normal ones;
- 2: Initialize model parameters by Glorot initialization [106].
- 3: **while** epoch $e < E$ **do**
- 4: Randomly sample N subgraphs from $\mathcal{C}^a$ and $\mathcal{C}^n$ respectively to form a training batch;
- 5: Estimate the anomaly score of each node through (1) and (4);
- 6: Calculate the objective function in (7);
- 7: Update model parameters by backpropagation;
- 8: Update $e \leftarrow e + 1$;
- 9: **end while**
- 10: Derive anomaly scores $\mathbf{c}$ and sort them in reverse order.

and abnormal subgraphs during the training process of ASSESS. Fig. 3 illustrates the node scores in these two subgraphs on different datasets, where the left column shows anomaly scores assigned by the model to nodes in a randomly selected ground-truth abnormal subgraph, and the right column shows node scores in a normal subgraph. As can be seen, there is a clear

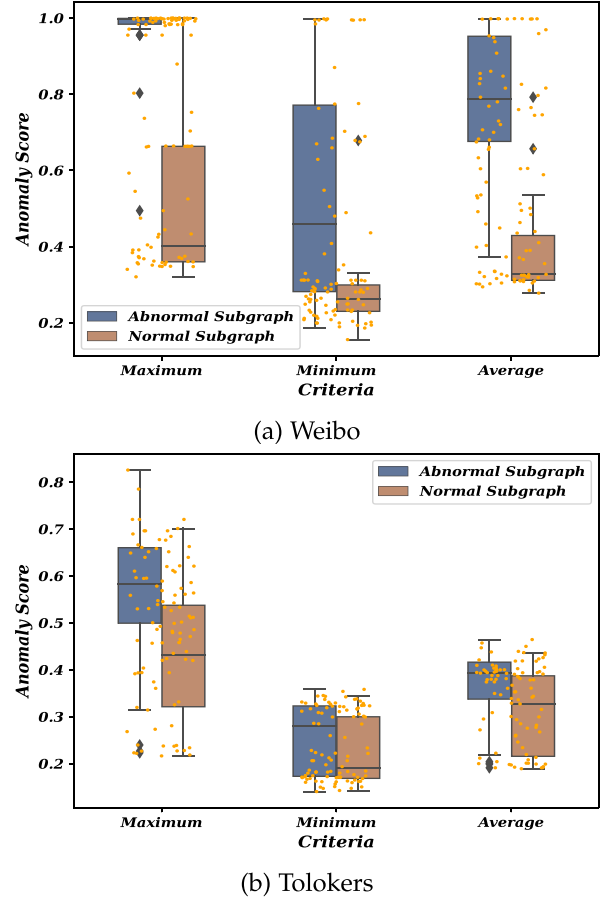


Fig. 4. Box plots of the maximum, minimum, and average scores of nodes in abnormal and normal subgraphs on Weibo and Tolokers.

difference in score distribution between these two types of subgraphs. The node scores in the normal subgraph are all below 0.5, while the abnormal subgraph has one or more nodes with scores above 0.5. Moreover, we find that the maximum, minimum, and average scores of nodes in abnormal and normal subgraphs have obvious statistical differences, as shown in Fig. 4. In this sense, measuring the score distribution of subgraphs seems to provide a good criterion for pseudo-annotation. Accordingly, we propose a simple yet effective strategy to automatically find unlabeled subgraphs that may contain anomalies to iteratively expand the label set $\mathcal{C}^a$. More sophisticated methods are left for future work (the difference in the number of nodes in each subgraph makes some commonly used distribution measurement methods such as KL divergence inapplicable). We first sample subgraphs with similar sizes to the labeled abnormal subgraphs for nodes in $\mathcal{V}^u$ to form a candidate subgraph set $\mathcal{C}^t = \{\mathcal{S}_1^t, \mathcal{S}_2^t, \dots, \mathcal{S}_w^t\}$. Then, the maximum, minimum, and average values are simultaneously used as indicators to roughly test the difference in score distribution between the candidate subgraph and the labeled sample. Formally, we score each candidate subgraph $\mathcal{S}_i^t$ by

$$f(\mathcal{S}_i^t) = \left| \frac{1}{k} \sum_{i=1}^k \max_{v_p^{(i)} \in \mathcal{S}_i^a} \mathbf{c}_p - \max_{v_j^{(i)} \in \mathcal{S}_i^t} \mathbf{c}_j \right|$$

$$\begin{aligned}
& + \left| \frac{1}{k} \sum_{i=1}^k \min_{v_p^{(i)} \in \mathcal{S}_i^a} c_p - \min_{v_j^{(i)} \in \mathcal{S}_i^t} c_j \right| \\
& + \left| \frac{1}{k} \sum_{i=1}^k \frac{1}{b_i} \sum_{v_p^{(i)} \in \mathcal{S}_i^a} c_p - \frac{1}{b_i^t} \sum_{v_j^{(i)} \in \mathcal{S}_i^t} c_j \right|, \quad (8)
\end{aligned}$$

where b_i^t denotes the size of $\mathcal{S}_i^t$. We claim that the smaller the difference between $\mathcal{S}_i^t$ and the samples in the label set $\mathcal{C}^a$ on the above three statistical indicators, the higher the possibility that it contains anomalies. Accordingly, we select the top r candidate subgraphs with the smallest scores and mark them with pseudo labels to expand $\mathcal{C}^a$ for the next training. Through multiple rounds of iterations, ASSESS++ automatically mines high-confidence training samples from unlabeled data and then fine-tunes itself with labeled and pseudo-labeled sets to achieve accurate node anomaly detection at low labeling rates.

C. Comparison of ASSESS and WEDGE

As mentioned above, MIL can be roughly divided into Bag-Space (BS) paradigm and Instance-Space (IS) paradigm [96]. Here we provide a formal discussion on the strengths and weaknesses of these two paradigms to help readers understand the superiority of our proposed method later. The BS paradigm treats each bag as a whole and learns the discriminative information at the bag-level. It typically assumes uniform relevance of instances within a bag, allowing the model to take into account global information by aggregating instance features to distinguish between different bags. By focusing on the bag as a whole, it is more robust to noise and irrelevant instances as the aggregation helps to mitigate the influence of outlier instances within a bag. However, BS paradigm may suffer from the above restrictive bias, and aggregating instances is prone to the loss of critical information, especially if the relevant patterns are only present at the instance level. This coarse-grained representation leads to limited expressiveness of the model. As for IS paradigm, it follows the standard multiple instance assumption that at least one instance in a positive bag is relevant to the positive annotation, which allows for effective learning and desirable generalization even with limited labeled data. The IS paradigm focuses on characterizing more detailed patterns of each instance and enables discrimination at the instance-level. This fine-grained representation facilitates capturing complex patterns that may be overlooked in an aggregated view, giving the model stronger expressiveness.

In addition to the above analysis, existing studies have shown theoretically and empirically that better performance in bag-level accuracy does not necessarily mean higher instance-level accuracy [95], [107], [108]. To briefly review, let TPR_i and TNR_i be the instance-level true positive/negative rate, on a set of L_i instances containing p_i positives and n_i negatives, the instance-level accuracy is calculated by

$$ACC_i = \frac{TPR_i \cdot p_i + TNR_i \cdot n_i}{L_i}. \quad (9)$$

Then on a set of L_b bags, where $\mathcal{B}^+$ and $\mathcal{B}^-$ contain positive and negative bags respectively, the bag-level accuracy is derived by

$$ACC_b = \frac{\sum_{B_k \in \mathcal{B}^+} a_b^+(B_k) + \sum_{B_k \in \mathcal{B}^-} a_b^-(B_k)}{L_b}, \quad (10)$$

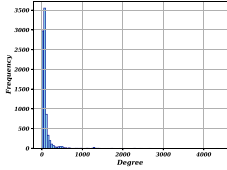
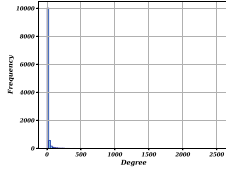
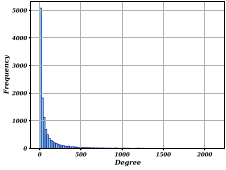
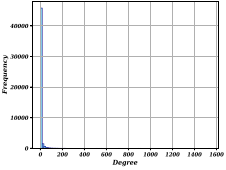
where $a_b^+(B_k) = 1 - (1 - TPR_i)^{p_k}$ and $a_b^-(B_k) = TNR_i^{n_k}$, p_k and n_k denote the numbers of positive and negative instances in bag B_k , respectively. As can be seen, when at least one instance (whether positive or negative) is predicted as a positive, the prediction of the positive bag is correct. In this sense, methods designed for bag classification are not optimal for accurately distinguishing instances since they are prone to learn a suboptimal decision boundary for instance classification. Besides, ACC_b is affected by many factors such as the size of the bags B_k , TPR_i , and TNR_i , ACC_i is more stable than ACC_b . For more discussion on the two types of paradigms, you could refer to [95], [107].

Next, we compare WEDGE with our work. Although both methods are based on MIL for weakly supervised node anomaly detection, they are significantly different in terms of technology and our well-designed self-training module improves the model's adaptability at low labeling rates. First, as the taxonomy introduced in § II-B, WEDGE adopts the BS paradigm where the discriminative information is considered to lie at the bag-level. WEDGE aggregates node (instance) embeddings and scores through an attention mechanism to obtain subgraph (bag) embeddings and scores. Then employs contrastive learning to establish discrimination between subgraphs. As analyzed above, the correct distinction of subgraphs (bags) does not mean the accurate judgment of the nodes (instances) in them. In contrast, we adopt the IS paradigm. Since the discriminative information is at the instance-level, our modeling strategy is more fine-grained. Specifically, we formulate anomaly detection as a regression problem. By ranking nodes with the maximum scores in normal and abnormal subgraphs, and pulling apart the difference between potential anomalies and normal nodes in the abnormal subgraph, ASSESS will assign high scores to anomalies to identify them. Second, note that many (weakly-)supervised methods including WEDGE fail to cope with low labeling rate scenarios (see experiments). To further enhance the model's adaptability, we introduce the self-training mechanism to form a model variant ASSESS++. By automatically mining subgraphs that may contain anomalies as subgraph-level pseudo labels, ASSESS++ expands the available label set iteratively, which significantly alleviates the high cost required to obtain high-quality annotations. Compared with WEDGE, our model has a wider range of practical application scenarios.

D. Time Complexity

The node representation modeling needs $O(n^2h + ndh + nh^2)$, the anomaly ranking takes $O(kp)$, the sparsity and Homophily constraints need $O(kp)$ and $O(kp^2)$ respectively, where h denotes the embedding dimension, k is the number of labeled subgraphs, and p denotes the average number of nodes contained in the labeled subgraph (mostly around 10). Since k and p are usually very small, the time complexity of ASSESS is

TABLE III
STATISTICS OF THE DATASETS USED IN THE EXPERIMENTS

	Weibo	Reddit	Tolokers	Questions
#Nodes	8,405	10,984	11,758	48,921
#Edges	407,963	168,016	519,000	153,540
#Features	400	64	10	301
Density	1.16%	0.28%	0.75%	0.01%
Degrees(max/min/avg)	4,445/1/97.08	2,555/1/14.29	2,138/1/88.28	1,539/1/6.28
Average Local Clustering	0.43	close to 0.00	0.53	0.03
Anomaly Ratio	10.3%	3.3%	21.8%	3.0%
Network Type	Tencent-Weibo Social Network	User-Subreddit Network	Collaboration Network	Question-Answering Network
Relation Formation	Under Same Hashtag	Under Same Post	Work Collaboration	Question Answering
Node Feature Type	Text Embedding	Text Embedding	Profile Information	Text Embedding
Degree Distribution				

$O(n^2h + ndh + nh^2)$, which has almost the same efficiency as those well-known graph representation methods such as DGI ($O(n^2h + ndh)$) [109]. Overall, our model processes at the subgraph granularity, which is a natural batch form that is advantageous for generalization to large-scale graphs.

IV. EXPERIMENTS

This section empirically evaluates the anomaly detection quality of the proposed framework. Moreover, we provide some important parameter analysis and visualize the detection results for intuitive understanding.

A. Datasets

We adopt the following four public real-world benchmark datasets [20] to validate the performance of the proposed AS-SESS and ASSESS++. Table III summarizes detailed statistics such as network size, structure, type, and anomaly ratio. The networks here are all homogeneous.³

- **Weibo** [110] is collected from the Tencent-Weibo online discussion platform, where nodes represent users and edges indicate that two users have the same hashtag. Users that engaged in at least five suspicious activities (*i.e.*, posting twice within a certain timeframe such as 60 seconds) are labeled as abnormal. Node features are composed of the location of a micro-blog post and bag-of-words vectors.
- **Reddit** [111] contains a user-subreddit graph, where nodes represent posts shared across the subreddits and edges between two posts indicate that they are commented by the same user. The text of each post is transformed into a feature vector representing their LIWC (Linguistic Inquiry and Word Count) categories. The banned users are labeled as anomalies.
- **Tolokers** [112] originates from the crowd-sourcing platform Toloka, where nodes are workers who have joined in at least one of the 13 selected projects. If two workers collaborate on the same task, they are linked by an edge.

The worker’s profile and task performance statistics form node features. Workers who have been banned from one of the projects are marked as abnormal.

- **Questions** [112] is collected from the question-answering website Yandex Q. Nodes represent users interested in the topic “medicine”, and links are established between two users if one user answers another’s question within a year. Users who remained active (not deleted or blocked) are marked as anomalies. Node features are the average of FastText embeddings of the words in the user description.

As can be seen, the above datasets differ in several network characteristics such as degree distribution, density, and anomaly ratio. In Questions, the node average degree and the average local clustering coefficient for the whole graph are both smaller, showing a more sparse topological structure. Although Weibo and Reddit are both online social networks, Weibo is relatively denser, while the clustering extent of neighbors around nodes in Reddit is close to 0 (low local clustering), indicating that there is almost no clique between neighbors (few users comment on multiple posts). In Tolokers, the higher local clustering coefficient indicates closer cooperation among workers. Based on the ground-truth node-level anomalies contained in these datasets, in this paper, we create $k = 50$ labeled abnormal subgraphs $\mathcal{S}^a$ as weak supervision. In detail, we first randomly select k true anomalies as center nodes, then generate local subgraphs via random walk with restart (RWR) [113] algorithm. The random walk length and the restart probability are set to 15 and 0.2, respectively. Similarly, the normal subgraph $\mathcal{S}^n$ is created by applying RWR on randomly chosen nodes from the unlabeled node set $\mathcal{V}^u$. In addition, we set $k = 10$ to simulate the low labeling rate situation.

B. Compared Methods

For a comprehensive comparison, we choose three categories of methods: unsupervised, supervised, and weakly supervised as baselines.⁴ The specific details are as follows.

³The proposed weakly supervised anomaly detection idea can be naturally extended to heterogeneous networks (by increasing the modeling of heterogeneous entities/relations). Related exploration is left for future research.

⁴Existing supervised methods depend on precise node-level labels. Strictly speaking, they cannot handle the case with only imprecise subgraph-level supervision, so we do not consider many such methods in our baselines.

- **ANEMONE** [62] is an *unsupervised* framework that employs multi-scale contrastive learning and a statistical anomaly estimation strategy to identify anomalous nodes.
- **GRADATE** [14] is an *unsupervised* multi-view multi-scale contrastive learning framework that introduces subgraph-subgraph contrast for node anomaly detection performance promotion.
- **RAND** [65] achieves *unsupervised* node anomaly detection with a tailored reinforcement neighborhood selection and anomaly-aware message aggregation.
- **GAD-NR** [68] presents a new variant of GAE that incorporates neighborhood reconstruction in terms of local structure, self-attributes, and neighbor attributes for *unsupervised* node anomaly detection.
- **ADA-GAD** [67] introduces a two-stage *unsupervised* framework that first trains the autoencoder on graphs with reduced anomaly levels generated by an anomaly-denoised augmentation strategy, then retrains the decoder for anomaly node detection on the original graph.
- **EAGLE** [70] disentangles nodes and attributes to reformulate the original graph into a special bipartite graph, then uncovers node anomalies in an *unsupervised* manner through bipartite graph learning.
- **GDN & Meta-GDN** [51] are GNN-based *few-shot* node anomaly detection approaches that enforce large deviations between abnormal and normal nodes by well-designed graph deviation networks, where Meta-GDN is additionally equipped with a new cross-network meta-learning algorithm.
- **BWGNN** [53] is a *supervised* architecture that introduces Beta graph wavelets to create band-pass filters with excellent locality for detecting node anomalies in the spectral domain.
- **SpaceGNN** [114] ensembles comprehensive information from different spaces with a specialized attention mechanism to spot node anomalies under limited node *supervised* labels.
- **MIL-GNN** [77] is a GNN-based MIL model that regards the graph as a bag to perform graph-level classification in a *weakly supervised* manner.
- **WEDGE** [23] is a *weakly supervised* architecture equipped with an attention module, which spots node anomalies by enhancing the similarity within subgraphs and the disparity between subgraphs.

Unsupervised methods identify anomalies in input graph data without available labels. As for supervised methods designed for node-level labels, following previous studies [23], all nodes in the labeled abnormal subgraphs act as supervisory signals. Regarding MIL-GNN specific for graph-level (bag-level) classification, we spot node anomalies by taking the predicted class probability of the subgraph as the anomaly score of the nodes within it.

C. Parameter Settings and Metrics

All experiments are implemented in PyTorch [115] and carried out on a single NVIDIA RTX 4090 GPU. Since WEDGE

does not have a public source code, we implemented it ourselves based on its published paper. All other baselines are implemented through their released source code and undergo parameter tuning according to the empirical strategies in their papers. As for our framework ASSESS and ASSESS++,⁵ we adopt vanilla 2-layer GCN as the graph encoder and Jensen-Shannon MI estimator (JSD) [116] for mutual information calculation. In specific, we derive $I(\mathbf{z}_i; \mathbf{x}_i)$ by

$$I(\mathbf{z}_i; \mathbf{x}_i) = -sp(-\mathcal{D}_w(\mathbf{z}_i, \mathbf{x}_i)) - \mathbb{E}_{\tilde{\mathbb{P}}}[sp(\mathcal{D}_w(\mathbf{z}_i, \mathbf{x}'_i))], \quad (11)$$

where $\mathcal{D}_w : d \times h_2 \rightarrow \mathbb{R}$ is a trainable discriminator with parameter w . $\mathbf{x}'_i$ is a negative sampled from $\tilde{\mathbb{P}} = \mathbb{P}$ (we denote the empirical probability distribution of node attributes $\mathbf{X}$ by $\mathbb{P}$), and $sp(x) = \log(1 + e^x)$ is the soft-plus function. The number of negative samples for the JSD estimator is set to 5. As for $I(w_{ij}; \mathbf{A}_{ij})$, we maximize it by calculating its cross-entropy instead of JSD estimator as the graph data used in experiments is unweighted:

$$I(w_{ij}; \mathbf{A}_{ij}) = \mathbf{A}_{ij} \log w_{ij} + (1 - \mathbf{A}_{ij}) \log(1 - w_{ij}). \quad (12)$$

During training, we employ RMSprop optimizer with the learning rate 0.001, the weight decay is set to 5×10^{-4} , and the batch size $N = 32$. We tune the three parameters λ s by a “grid-search” strategy in the range of [0.01, 1]. About ASSESS++, we create $r = 5$ pseudo labels every 100 epochs until a total of 50 labeled abnormal subgraphs are reached.

Following the commonly used metrics in the previous studies [14], [17], we employ the criteria of AUC, which is the area under the ROC curve, to evaluate the detection quality of various methods. Besides, the AP score (Average Precision) is also utilized to evaluate the ability of the model to identify positive instances accurately while maintaining high precision. AP is more suitable for anomaly detection tasks where positive (abnormal) and negative (benign) samples are extremely imbalanced. The closer the AUC and AP scores approach 1, the better the detection performance of the method is. In this paper, we report the average results with standard deviation over five runs.

D. Main Experimental Results

We summarize the evaluation results under different labeling ratios in Table IV and have the following observations. **In the weakly supervised case with $k = 50$ subgraph-level labels**, our proposed ASSESS consistently outperforms all baselines in terms of AUC and AP scores on four datasets. The performance gains compared to existing weakly supervised models MIL-GNN and WEDGE reflects the effectiveness of the ranking-based weakly supervised detection framework we designed. Especially compared to WEDGE based on the Bag-Space paradigm, which establishes discrimination between subgraphs (bag-level), the Instance-Space paradigm we adopt is more fine-grained. By ranking nodes with the maximum scores in normal and abnormal subgraphs, and pulling apart the difference between potential anomalies and normal nodes in the abnormal subgraph, we achieve the discrimination at the

⁵Source code is available at <https://github.com/zpeng27/ASSESS>.

TABLE IV
PERFORMANCE OF DIFFERENT METHODS W.R.T. AUC AND AP (%) ON REAL-WORLD DATASETS

<i>k=50 subgraph-level labels</i>									
ALG.		Weibo		Reddit		Tolokers		Questions	
		AUC	AP	AUC	AP	AUC	AP	AUC	AP
unsupervised	ANEMONE	20.37±1.46	9.78±1.56	55.77±0.24	3.91±0.04	56.78±0.31	24.97±0.17	44.88±0.47	2.67±0.03
	GRADATE	21.98±1.14	6.83±0.14	56.33±0.91	3.79±0.18	52.80±1.18	23.26±0.45	47.96±0.80	2.94±0.09
	RAND	23.72±0.27	6.34±0.18	56.37±0.11	3.78±0.01	55.12±0.16	28.39±0.14	61.72±0.28	4.63±0.11
	GAD-NR	73.27±2.27	15.44±1.47	58.35±1.35	4.47±0.17	57.38±2.58	25.78±1.25	59.11±1.11	4.16±0.81
	ADA-GAD	90.36±1.25	64.84±0.53	56.88±1.42	4.05±0.13	56.21±1.17	23.73±0.85	OOM	OOM
	EAGLE	88.57±0.87	52.34±2.15	58.12±0.12	4.08±0.06	61.13±1.86	26.84±1.15	60.67±0.21	4.53±0.03
supervised	GDN	68.02±2.16	35.56±3.07	53.42±1.81	3.75±0.62	55.45±2.01	24.96±2.86	53.29±0.85	2.71±1.54
	Meta-GDN	69.37±2.68	37.65±1.54	58.51±2.64	4.36±0.52	59.53±3.06	29.52±2.98	55.78±0.93	3.79±0.34
	BWGNN	70.26±0.49	46.92±0.24	62.54±0.33	4.71±0.03	66.83±0.15	35.35±0.49	52.19±0.21	3.72±0.09
	SpaceGNN	87.17±1.53	47.79±0.64	63.08±0.63	5.50±0.26	67.81±0.19	36.15±1.49	65.89±0.58	6.15±0.26
weakly-supervised	MIL-GNN	93.52±2.24	77.33±3.30	56.01±1.85	3.77±0.11	59.47±1.49	31.30±0.96	58.89±1.79	4.17±0.26
	WEDGE	97.87±0.18	92.01±0.59	63.95±0.09	5.63±0.23	68.92±0.59	38.71±0.86	61.87±0.34	4.41±0.05
	ASSESS	98.60±0.11	93.08±0.42	67.11±1.37	5.92±0.38	70.10±0.45	39.87±0.32	67.47±0.72	6.84±0.16
<i>k=10 subgraph-level labels</i>									
supervised	GDN	52.43±3.01	16.81±1.79	49.42±0.73	2.43±0.45	53.33±1.22	15.85±1.71	48.96±0.43	2.95±0.61
	Meta-GDN	60.86±4.95	27.05±2.95	52.62±0.89	3.03±0.17	55.59±4.53	25.66±3.24	51.41±0.67	3.26±0.30
	BWGNN	51.81±1.05	42.91±2.16	57.17±1.53	4.43±0.62	62.69±0.52	33.73±0.48	45.56±0.44	2.99±0.13
	SpaceGNN	79.32±2.06	41.29±1.36	59.25±0.83	4.86±0.17	58.09±0.46	25.84±1.72	63.77±0.61	5.62±0.45
weakly-supervised	MIL-GNN	76.04±1.37	58.81±2.21	50.57±2.05	3.13±0.11	52.53±1.82	26.84±1.27	52.61±1.61	3.34±0.25
	WEDGE	96.98±0.61	89.27±0.69	55.31±1.73	4.39±0.35	66.63±1.59	35.15±0.79	56.59±0.59	3.79±0.07
	ASSESS++	97.22±0.60	89.90±0.72	64.77±0.74	5.68±0.29	69.39±1.02	36.31±0.51	66.29±0.31	6.61±0.24

The best results are marked in bold.

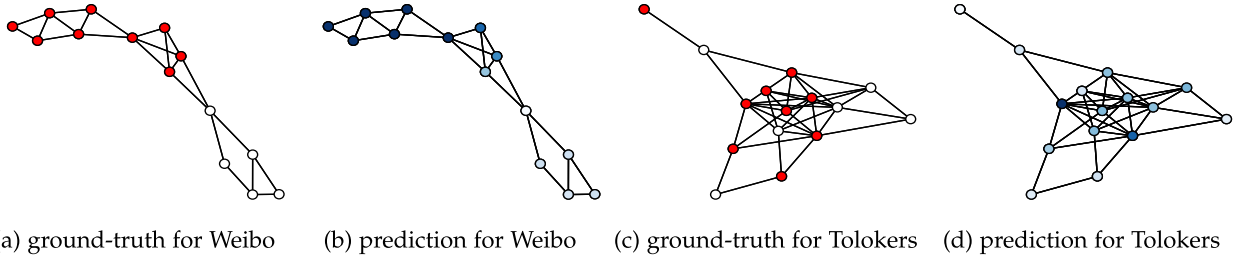


Fig. 5. Visualization of node anomaly scores in a randomly chosen abnormal subgraph predicted by ASSESS. The darker the color, the greater the score, and the more likely the node is an anomaly.

node-level (instance-level), which demonstrates superior detection performance. Additionally, the significant performance gap between our method and unsupervised methods such as EAGLE and GAD-NR illustrates the benefit of introducing external label knowledge as guidance, even with coarse-grained supervision. Since the autoencoder-based model ADA-GAD is trained to reconstruct the network topology and attributes of multiple additional augmented graphs, which takes up a lot of storage space, it faces a serious OOM (out of memory) problem on large-scale network datasets such as Questions. As for supervised or even few-shot learning methods that rely on node-level labeling, they can only extract limited or even incorrect knowledge from the coarse-grained supervisory signals, thus weakening the performance. **When k is reduced to 10 to reach a relatively low labeling rate condition**, our ASSESS++ consistently achieves the best across all datasets. Compared to other baselines with obvious performance degradation, we get desirable results that are almost comparable to those when $k = 50$. This is due to the simple but effective self-training module we designed, which automatically creates subgraph-level pseudo-labels by

measuring the score distribution of nodes to enhance the model's adaptability to low labeling rates in reality. Since unsupervised methods do not rely on labels and the results remain unchanged, it is omitted here.

In addition to quantitative results, we also provide some visualizations of the detection results output by ASSESS to intuitively demonstrate its good performance. Specifically, we randomly select an abnormal subgraph from Weibo and Tolokers datasets respectively, where the ground-truth abnormal nodes are marked as red. Then, the node anomalies identified by our ASSESS are colored blue. The darker the color, the greater the anomaly score and the higher the possibility of an anomaly. By comparing the subfigures (a) and (b) or (c) and (d) in Fig. 5, it can be seen that most real anomalies were successfully assigned high scores by ASSESS, although there were a few misjudgments. Furthermore, on Weibo dataset, we summarize the anomaly scores predicted by different models for the two types of nodes, which are categorized according to the ground-truth abnormal and normal labels, and illustrate violin plots as shown in Fig. 6. Obviously, ASSESS assigns higher anomaly scores to true node

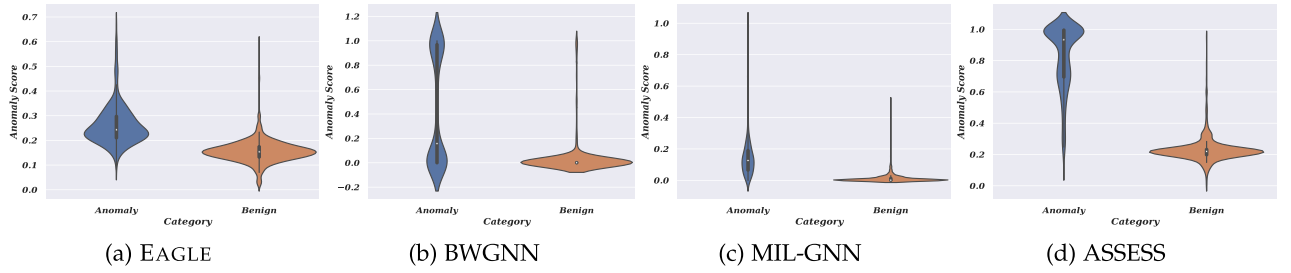


Fig. 6. Violin plots of node scores predicted by different models on Weibo dataset, grouped by true anomalies and benign.

TABLE V
PERFORMANCE UNDER DIFFERENT GRAPH ENCODERS

Data		GCN	GIN	GAT	MixHop
Weibo	AUC	98.60±0.11	97.87±0.24	99.04±0.19	98.97±0.63
	AP	93.08±1.02	92.17±0.98	93.85±0.61	94.02±0.81
Reddit	AUC	67.11±1.37	65.14±0.68	65.37±1.14	65.48±1.27
	AP	5.92±0.38	6.08±0.19	5.90±0.22	6.19±0.39
Tolokers	AUC	70.10±0.45	70.04±0.36	70.53±0.49	71.25±1.91
	AP	39.87±0.32	39.74±0.08	39.91±0.15	40.28±0.95
Questions	AUC	67.47±0.72	68.14±0.25	68.59±0.51	68.62±1.13
	AP	6.84±0.16	6.92±0.18	7.01±0.36	8.38±0.56

anomalies, and there is a significant gap between the median scores of abnormal and normal nodes, forming a clear distinction boundary. In other methods, however, the two types of nodes have many overlapping score intervals, and many true anomalies are assigned scores similar to normal nodes, which leads to unsatisfactory detection effects. The above visualization results once again confirm the effectiveness of ASSESS.

E. More Experimental Analysis

1) *Adaptability to Different Graph Encoders*: In this part, we intend to explore the adaptability of our proposed ASSESS to different backbone GNN encoders. The well-known GAT [99], GIN [100], and MixHop [117] models are chosen as alternative encoders and we keep the other experimental settings unchanged. Table V summarizes the results. As can be seen, these encoder variants all exhibit comparable detection performance to GCN on different datasets. Note that GAT adaptively reweights neighborhoods through the self-attention mechanism, and MixHop captures neighborhood mixing relationships at various distances by multiple powers of the adjacency matrix. This fine-scale modeling of graph topology information is conducive to identifying anomalies, thus they further improves the values of both evaluation metrics on Weibo, Tolokers, and Questions. These promising and comparable results demonstrate that our model allows for a wide selection of backbone GNN encoders.

2) *Subgraph Generation Strategies*: Here we explore the impact of weak supervision signals created by different subgraph (bag) generation strategies on model performance. In addition to the random walk with restart (RWR), we also consider two additional methods: vanilla random walk (RW) and breadth-first random walk (BFRW). It needs to be recognized that

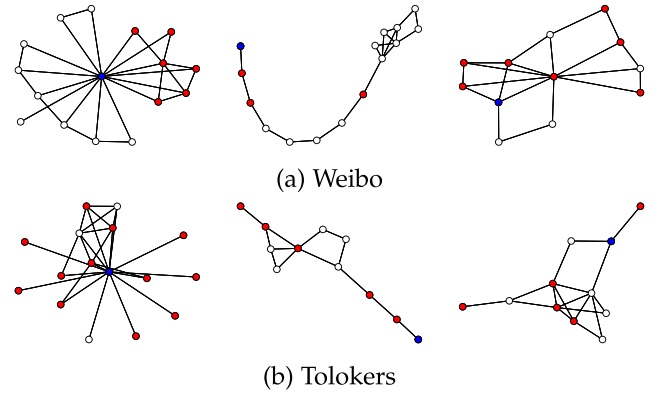


Fig. 7. Subgraphs derived by different strategies on Weibo and Tolokers: BFRW (left), RW (middle), and RWR (right). The abnormal nodes that serve as starting nodes for sampling are colored by blue, and the true node anomalies involved in the subgraph are marked in red.

TABLE VI
PERFORMANCE UNDER DIFFERENT SUBGRAPH GENERATION STRATEGIES

Data		RWR	RW	BFRW
Weibo	AUC	98.60±0.11	98.46±0.45	98.70±0.08
	AP	93.08±1.02	93.41±1.46	93.33±0.94
Reddit	AUC	67.11±1.37	67.72±0.08	65.76±0.62
	AP	5.92±0.38	5.94±0.14	5.93±0.15
Tolokers	AUC	70.10±0.45	70.08±0.43	72.20±0.12
	AP	39.87±0.32	39.78±0.19	40.19±0.14
Questions	AUC	67.47±0.72	68.37±0.81	67.20±0.56
	AP	6.84±0.16	7.40±0.46	6.81±0.39

the subgraphs derived by the above strategies have their own characteristics. Specifically, since RWR introduces the restart probability, *i.e.*, returning to the starting node, it tends to generate a subgraph with the abnormal node as the center, containing neighbors with diverse hops away from the abnormal node. In contrast, the vanilla RW is more likely to create a subgraph that exists in a chain-like form, starting from the abnormal node. For the BFRW, it is prone to generating a subgraph centered around the abnormal node, containing all neighbors with the same hops from the abnormal node. Fig. 7 shows some subgraphs derived by these strategies, and Table VI reports empirical results on four datasets. As can be seen, the performances under the above strategies are comparable, indicating that the proposed model

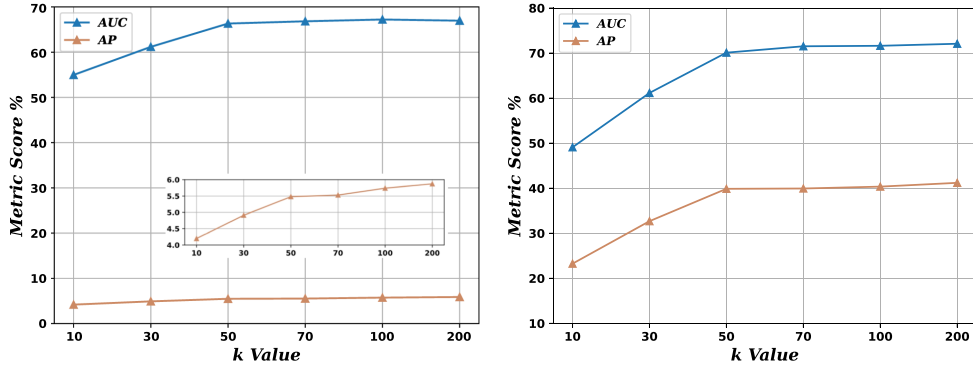


Fig. 8. Metric scores on Reddit (left) and Tolokers (right) under different numbers of labeled abnormal subgraphs.

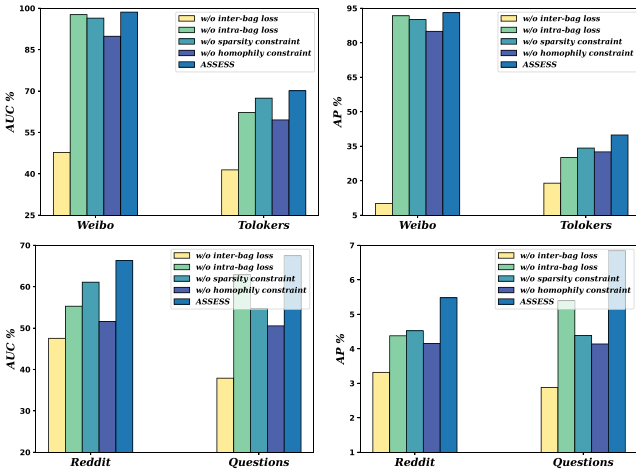


Fig. 9. Ablation studies on real-world datasets w.r.t. AUC and AP metrics.

can adapt to a variety of subgraph (bag) generation methods and has promising anomaly detection capability under different forms of subgraph-level weak supervision.

3) *The Influence of K Value*: For weakly supervised anomaly detection, it is interesting to explore the performance changes of ASSESS under different amounts of weak supervision. Fig. 8 illustrates the detection results of ASSESS as the number of annotated abnormal subgraphs (*i.e.*, k value) changes. When k is less than 50, both AUC and AP increase significantly, while the performance gain is only a little and almost remains stable when k continues to increase (further performance gains may come from more sophisticated model designs, which are left for the future). This reflects the necessity of sufficient weak supervisory signals, which motivates us to propose ASSESS++ that can automatically generate pseudo labels to further alleviate the labeling cost.

4) *Ablation Study*: We examine the role of each module in weakly supervised anomaly ranking modeling by removing the specific loss terms. In detail, w/o inter-bag loss, w/o intra-bag loss, w/o sparsity constraint, and w/o Homophily constraint refer to removing the $\ell(\mathcal{S}_i^a, \mathcal{S}_i^n)$, $\ell(\mathcal{S}_i^a)$, $\ell_{sp}(\mathcal{S}_i^a)$, and $\ell_{hom}(\mathcal{S}_i)$, respectively. Fig. 9 illustrates the results under different settings on all datasets. As can be seen, both AUC and AP metrics

drop most significantly when the inter-bag loss is missing, which indicates that the inter-bag anomaly ranking is critically important as it broadly captures the explicit differences between anomaly and benign. The performance degradation without $\ell(\mathcal{S}_i^a)$ reflects the benefit of modeling intra-bag ranking loss in further distinguishing normal nodes in coarse-grained abnormal subgraphs. In addition, modeling the sparsity of anomalies and the homophily principle inherent in graph topology are both beneficial to identifying anomalies accurately.

5) *Computational Complexity*: Here we evaluate ASSESS and other baselines in terms of the computational efficiency and number of model parameters under the same hardware environment. Table VII summarizes the results, where #Params indicate the number of trainable parameters that a machine learning model adjusts during training to improve its performance, and MACs denote multiply-accumulate operations⁶. The lower MACs value means that the model requires less computation and is more efficient. It can be seen that, compared to WEDGE, which is a representative weakly supervised model specifically for node anomaly detection and has the second-best performance, our proposed ASSESS can achieve the best detection performance with fewer parameters and computational burden. Compared to others, ASSESS has a relatively moderate number of parameters, similar to the widely used ANEMONE on all four datasets. For the pseudo-labeling stage in ASSESS++, additional dynamic memory allocation of 5.59 M, 6.58 M, 7.29 M, and 26.41 M is consumed on the four datasets respectively. Although larger MACs are consumed, this contributes to the sophisticated utilization and analysis of coarse-grained subgraph-level labels to accommodate practical weakly supervised scenarios. In addition, we analyze the detection duration of the model on datasets of different scales, as shown in Fig. 10. Here Elliptic dataset [118] includes a graph of 203,769 Bitcoin transactions as nodes, and 234,355 payment flows as edges, where each node has 166-dimensional transaction information and 9.8% transactions belongs to illicit categories such as scams and Ponzi schemes. On such a large-scale network, most methods such as RAND, ADA-GAD, EAGLE, GDN, and BWGNN suffer from the OOM issue. ANEMONE and GRADATE achieve an AUC score of

⁶The well-known FLOPs are nearly two times as MACs, more details at <https://github.com/Lyken17/pytorch-OpCounter>.

TABLE VII
THE COMPUTATIONAL COMPLEXITY OF DIFFERENT ALGORITHMS ON REAL-WORLD DATASETS

ALG.		Weibo		Reddit		Tolokers		Questions	
		#Params	MACs	#Params	MACs	#Params	MACs	#Params	MACs
unsupervised	ANEMONE	0.205M	246.374M	0.004M	4.992M	0.001M	0.845M	0.077M	92.774M
	GRADATE	0.409M	492.749M	0.008M	9.984M	0.001M	1.689M	0.154M	185.549M
	RAND	0.336M	2.823G	0.006M	67.485M	0.003M	31.606M	0.110M	5.373G
	GAD-NR	0.012M	930.207M	0.007M	1.156G	0.006M	1.228G	0.010M	5.337G
	ADA-GAD	0.197M	1.652G	0.003M	33.743M	0.003M	36.120M	OOM	OOM
	EAGLE	0.642M	3.447G	0.018M	114.966M	0.012M	79.961M	0.223M	6.520G
supervised	GDN	0.103M	345.129M	0.002M	9.137M	0.001M	1.655M	0.038M	756.421M
	BWGNN	0.496M	4.170G	0.008M	90.684M	0.006M	76.756M	0.137M	6.706G
	SpaceGNN	0.037M	520.022M	0.004M	8.517M	0.001M	10.303M	0.018M	55.941M
weakly-supervised	MIL-GNN	0.169M	151.984M	0.003M	2.467M	0.001M	1.204M	0.055M	40.447M
	WEDGE	0.399M	3.374G	0.184M	2.035G	0.183M	2.158G	0.269M	13.275G
	ASSESS	0.201M	1.695G	0.004M	45.254M	0.002M	24.127M	0.063M	3.112G

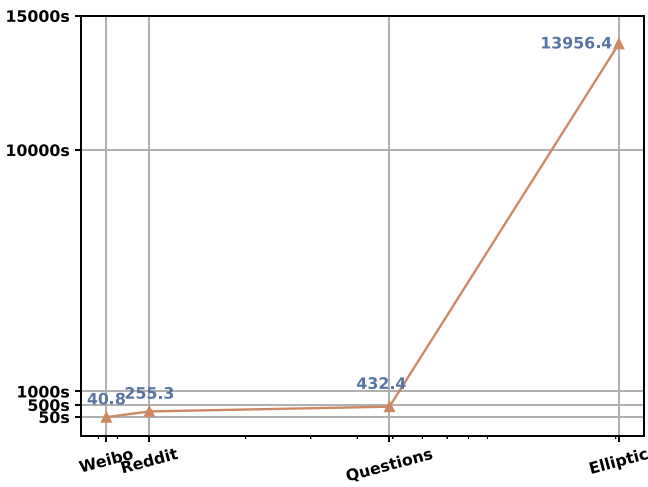


Fig. 10. Detection duration on datasets of different scales.

about 43.28%, but they take more than 24 hours (1 d) to complete. In contrast, our model only takes about 4 hours (13956.4 s) to achieve an AUC score of 86.31%. WEDGE only gets 64.83% in a similar period of time. Overall, our method exhibits excellent effectiveness and efficiency in weakly supervised node anomaly detection, reflecting good engineering application potential.

V. CONCLUSION

This paper investigates how to locate node-level anomalies under imprecise subgraph-level supervision, which is widely applicable in practice yet has received less scrutiny. Inspired by classic MIL methods, we treat subgraphs and nodes as bags and instances, respectively, and present a novel framework ASSESS which enforces inter-bag and intra-bag ranking loss to learn node abnormalities. A variant ASSESS++ equipped with a self-training module is proposed to further enhance the adaptability to low labeling ratios. Experiments on four benchmark real-world datasets validate the superiority of ASSESS and ASSESS++.

Future work could focus on the following three aspects. (1) Note that the proposed MIL-based weakly supervised node

anomaly detection idea is independent of the specific model architecture. Our work can be naturally extended to dynamic networks by introducing a dynamic graph representation learning module. For instance, employing DDGCL [119] or CLDG [120] derives node representations on dynamic graphs in a self-supervised manner, and then learn a temporal score predictor to model node anomaly ranking under the guidance of subgraph-level weak supervision. (2) In addition, designing more sophisticated subgraph distribution measures to create high-confidence pseudo-labels automatically or drawing on other high-quality pseudo-label generation methods [121] to further enhance the performance of the model at low labeling rates is also a promising future direction. (3) Note that there are inevitably camouflaged anomalies disguising themselves to evade detection in reality, which may blur the pattern difference between anomalies and benign ones. Existing empirical studies usually divide camouflages into two types: the feature camouflage and the relation camouflage [49]. In this regard, it is possible to consider introducing relevant feature screening mechanisms or employing reinforcement learning (RL) to select optimal neighbors and high-order topological relations to overcome the representation similarity issue that may be caused by vanilla GNN aggregation. This will help the model capture the subtle boundary between normal and abnormal instances, improving the ability to detect camouflaged anomalies.

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